

The Nagata Embedding Theorem

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Abstract

Except for a few corrections added in 2010, this article reproduces notes sent to Nagata in the 1970s, translating into the language of schemes his papers [2] and [3]. The proofs are a “constructive” version of Nagata’s proofs. Making the proofs constructive made it possible to avoid the use of valuations. A more detailed exposition and additional results are given in Conrad [1]. Unless the contrary is expressly stated, all schemes considered below are assumed noetherian.

1 Recalls on blow-ups

0.1

Let $\mathfrak{a}$ be an ideal of the ring A . The scheme obtained from $X = \text{Spec}(A)$ by blowing up $\mathfrak{a}$ is the union of the open subschemes $D(x)$, where x runs through a system of generators of $\mathfrak{a}$, and $D(x)$ is the spectrum of the subring $A[\mathfrak{a}x^{-1}]$ of $A[x^{-1}]$. Moreover,

$$D(x) \cap (X - V(\mathfrak{a})) = \text{Spec}(A[x^{-1}]).$$

If $\mathfrak{c}$ is a second ideal of A , defining a closed subscheme $V(\mathfrak{c})$ of X , the trace on $D(x)$ of the strict transform of $V(\mathfrak{c})$ is defined by the ideal of $A[\mathfrak{a}x^{-1}]$ which is the trace of the ideal of $A[x^{-1}]$ generated by $\mathfrak{c}$.

Lemma (0.2). *Let X be a scheme and let $\mathfrak{a}$ and $\mathfrak{b}$ be two sheaves of ideals on X . Let $\widehat{X}$ be the blow-up of X along $\mathfrak{a} + \mathfrak{b}$. Then the open subset of $\widehat{X}$ which is the union of the $D(x)$ for $x \in \mathfrak{b}$ contains the strict transform of $V(\mathfrak{a})$.*

Proof. The scheme $\widehat{X}$ is the union of the $D(x)$ for $x \in \mathfrak{a}$ or $x \in \mathfrak{b}$, and, for $x \in \mathfrak{a}$, the open subset $D(x)$ is disjoint from the strict transform of $V(\mathfrak{a})$. $\square$

The lemma 0.3 of the original text was false. The corrected version of the lemma which follows dates from March 1996. It forced a change of notation. The original lemma used the following notation, to which the rest of the text refers: instead of F and G , one considered their complements U and V , which, with $Y' := U \cap p^{-1}(Y)$, formed a commutative diagram

$$\begin{array}{ccc} Y' & \longrightarrow & Z \\ \downarrow & & \downarrow p \\ U & \longrightarrow & X, \end{array}$$

where Y' is closed in $p^{-1}(V)$.

Lemma (0.3). *Let $p : Z \rightarrow X$ be a morphism, let $F \supset G$ be closed subschemes of X , and let Y be a closed subscheme of Z . Suppose that*

$$Y \cap p^{-1}(F) \subset p^{-1}(G)$$

scheme-theoretically. Blow up X along G , obtaining $\widehat{X}$, and let $\tilde{p} : \widehat{Z} \rightarrow \widehat{X}$ be obtained from $p : Z \rightarrow X$ by base change. Then the closure of $Y - p^{-1}(G)$ in $\widehat{Z}$ and $\tilde{p}^{-1}(F - G)$, taken in $\widehat{X}$, are disjoint closed subschemes of $\widehat{Z}$.

Proof. We may suppose that Z and X are affine. Let $\mathfrak{G}$ be the ideal of G . Cover $\widehat{X}$ by the open subsets $D(g)$, for $g \in \mathfrak{G}$ (a system of generators of $\mathfrak{G}$ suffices), and cover $X - G$ by

$$U(g) = D(g) - (\text{inverse image of } G).$$

Thus

$$\begin{array}{ccc} X - G & \hookrightarrow & \widehat{X} \\ \uparrow & & \uparrow \\ U(g) & \hookrightarrow & D(g). \end{array}$$

It is enough to see that, for each g , the traces on $\tilde{p}^{-1}D(g)$ of $\tilde{p}^{-1}(F - G)$ and of the closure of $Y - p^{-1}G$ are disjoint. They are respectively the inverse image of the closure of $F \cap U(g)$ in $D(g)$, and the closure of $Y \cap p^{-1}(U(g))$ in $\tilde{p}^{-1}D(g)$.

Let a tilde denote inverse image, either from X to Z , or from $D(g)$ to $\tilde{p}^{-1}D(g)$. Since $p^{-1}G \supset Y \cap p^{-1}(F)$, the inverse image $\tilde{g}$ of g on Z can be written

$$\tilde{g} = y + f,$$

where y vanishes on Y and f vanishes on $p^{-1}(F)$. There are functions a_i on Z and elements f_i in the ideal of F such that

$$f = \sum_i a_i \tilde{f}_i.$$

Since $F \supset G$, the f_i are in the ideal of G , and the functions f_i/g are regular on $D(g)$. They vanish on $F \cap U(g)$, hence on the closure of $F \cap U(g)$ in $D(g)$, and therefore

$$\sum_i a_i (f_i/g)^\sim$$

vanishes on $\tilde{p}^{-1}\overline{F \cap U(g)}$. On $p^{-1}U(g)$ one has

$$1 = y/g + f/g = y/g + \sum_i a_i (f_i/g)^\sim,$$

and the same sum is therefore equal to 1 on $Y \cap p^{-1}U(g)$, hence on its closure in $\tilde{p}^{-1}D(g)$. This proves the required disjointness. $\square$

Remark. Suppose instead that one is given closed subschemes $F \supset G_0$ of X , with $Y \cap p^{-1}(F)$ contained set-theoretically in $p^{-1}(G_0)$. Then there exists an r such that the r -th infinitesimal neighbourhood G of G_0 in F satisfies the hypotheses of the lemma. If J is the ideal defining $(Y \cap p^{-1}(F))_{\text{red}}$ in $Y \cap p^{-1}(F)$, it is enough to take r such that $J^{r+1} = 0$. Thus the lemma permits, by a blow-up which does not touch the complement V of G_0 , the separation of the strict transform of F from the closure of $Y - p^{-1}(G_0)$ in $\widehat{Z}$.

Lemma (0.4). Let

$$\begin{array}{ccc} X' & \xrightarrow{p} & X' \\ q \downarrow & & \downarrow \\ U & \hookrightarrow & X \end{array}$$

be a commutative diagram of schemes, with U a dense open subset of X and of X' , with p and q separated of finite type, and with $q^{-1}(X) = X'$. Let F be a closed subscheme of X , and let G be a closed subscheme of X' containing $p^{-1}(F)$. Then there exists an ideal $\mathfrak{a}$, with $V(\mathfrak{a}) \subset F - \overline{F}$, such that, after base change by the blow-up morphism defined by $\mathfrak{a}$, $\widehat{X} \rightarrow X$, and after replacing the new X' by the closure of U , one has

$$q^{-1}(F) \subset G.$$

Proof. In Nagata's language, to take the closure of F in the Zariski–Riemann space of X , it is enough to take the “intersection” of the closures of F in the blow-ups of X whose centres are contained in $F - \overline{F}$.

Apply Lemma 0.3 to

$$\begin{array}{ccc} X' - q^*F & \longrightarrow & X' - G \\ \uparrow & & \uparrow \\ X - F & \longrightarrow & X - (F - \overline{F}). \end{array}$$

One obtains $\mathfrak{a}$, and, after the base change $\widehat{X} \rightarrow X$, the new closure of F is disjoint from the image of the closure of the old $X' - q^*F$ in $X' - G$. In particular, after replacing $\widehat{X}$ by U , this new closure is disjoint from the image of $X' - G$, which is what was required. $\square$

Lemma (0.5). *Let*

$$\begin{array}{ccc} X_i & \longrightarrow & X_i \quad (1 \leq i \leq n) \\ \uparrow & & \uparrow \\ U & \longrightarrow & X \end{array}$$

be a diagram of separated S -schemes of finite type, with X_i proper over S , and with U a dense open subscheme of X and of the X_i . Let X^ be the closure of U in the product of the X_i . Finally, let F_i be a closed subscheme of X_i ; suppose that the intersection of the inverse images of the F_i in X^* is empty. Then, after replacing the X_i by blow-ups whose centres lie in $F_i - \overline{F_i}$, one obtains an analogous situation in which the intersection of the inverse images of the $\overline{F_i}$ is empty.*

Proof. Let X^{**} be a blow-up of X^* , with projections $p_i : X^{**} \rightarrow X_i$, such that in X^{**} one has

$$\bigcap_i p_i^{-1}(F_i) = \emptyset;$$

for example, blow up $\bigcap_i p_i^{-1}(F_i)$. Apply Lemma 0.4 to the diagrams

$$\begin{array}{ccc} p_i^{-1}(X_i) & \longrightarrow & X^{**} \\ \uparrow & & \uparrow \\ U & \longrightarrow & X_i, \end{array}$$

and to $F_i \subset X_i$, $G_i = p_i^{-1}(F_i)$. This provides the desired blow-ups of the X_i . Indeed, in the new situation, in the closure of U in X^{**} , one has

$$\bigcap_i p_i^{-1}(\overline{F_i}) \subset \bigcap_i G_i = \emptyset.$$

Since $X^{**} \rightarrow X^*$ is surjective, this remains true in X^* . $\square$

2 The theorems

Definition (1.1). *Let X and Y be two S -schemes, with Y separated over S . A quasi-domination $f : X \dashrightarrow Y$ is a pair consisting of a dense open subscheme U of X and a morphism of S -schemes $f : U \rightarrow Y$ whose graph is closed in $X \times_S Y$.*

Let X and Y be two S -schemes, with Y separated over S , let U be a quasi-compact open subscheme of X , and let $f : U \rightarrow Y$. We say that X is quasi-dominant over Y , relative to U and f , if there exists a quasi-domination of the schematic closure $\overline{U}$ of U , with values in Y , which extends f . There is at most one such quasi-domination; its graph is the closure of the graph of f .

A quasi-domination f is called proper or complete if the morphism $f : U \rightarrow Y$ is proper.

Theorem (1.2). *Let X and Y be two S -schemes, and let $U \subset V \subset X$ be two dense open subschemes of X . Suppose that Y is separated of finite type over S . Let $f : U \rightarrow Y$ be a quasi-dominion of V in Y . Then there exists an ideal $\mathfrak{a}$, with $V(\mathfrak{a}) \subset X - V$, such that the blow-up $\widehat{X}$ of X with centre $V(\mathfrak{a})$ quasi-dominates Y .*

Proof. By the base change $X \rightarrow S$ one is reduced to the case $X = S$. Then Y is a scheme over X , f is a section of Y over U , and by hypothesis $f(U)$ is closed over V .

Case 1: $U = V$, $p : Y \rightarrow X$ quasi-affine, and X affine. By hypothesis there is a commutative diagram

$$\begin{array}{ccc} Y & \longrightarrow & E_X \\ & \searrow & \downarrow \\ & & X, \end{array}$$

where E_X is an affine space over X . If $\widehat{X}$ quasi-dominates E_X relative to f , then a fortiori $\widehat{X}$ quasi-dominates Y . We may therefore suppose $Y = E_X$. A preliminary blow-up allows us to suppose that $X - U$ is a divisor D . Locally on X , the section f has coordinates $(f_i u^{-n})$, where u is a local equation of D , where the f_i are regular, and where n is chosen once and for all sufficiently large. Let $\mathfrak{a}$ be the ideal generated by u^n and the f_i . After blowing up $V(\mathfrak{a})$, the section f extends to a section of the projective completion of E_X , and we are done.

Case 2: $U = V$ and p quasi-affine. By Case 1, if $(U_i)_{0 \leq i \leq n}$ is an affine open covering of X , there exists on U_i an ideal $\mathfrak{a}_i$ which works. This ideal admits an extension $\tilde{\mathfrak{a}}_i$ to X , with $V(\tilde{\mathfrak{a}}_i) \subset X - V$, and it is enough to blow up the product of the $\tilde{\mathfrak{a}}_i$.

Case 3: p quasi-affine. By Lemma 0.3, applied to

$$\begin{array}{ccc} f(U) & \longrightarrow & Y \\ & & \downarrow p \\ U & \hookrightarrow & V \hookrightarrow X, \end{array}$$

a preliminary blow-up allows us to suppose that

$$p(f(U)) \cap (V - U) = \emptyset.$$

There is therefore a quasi-compact open subscheme X' of X with $X' \cap V = U$ and $p(f(U)) \subset X'$. By Case 2, there exists an ideal $\mathfrak{a}$ on X' , with $V(\mathfrak{a}) \subset X' - U$, such that the blow-up $\widehat{X'}$ quasi-dominates Y , that is, $f(U)$. It remains only to extend $\mathfrak{a}$ to an ideal on X , with $V(\mathfrak{a}) \subset X - V$.

General case. Let (Y_i) be a finite covering of Y by affine open subschemes, and put

$$U_i = f^{-1}(Y_i), \quad X_i = \overline{U_i}, \quad V_i = X_i \cap V.$$

Case 3 applies to

$$Y_i \cap p^{-1}(X_i), \quad U_i \subset V_i \subset X_i,$$

and gives an ideal $\mathfrak{a}_i$ on X_i . This ideal defines an ideal on X , still denoted $\mathfrak{a}_i$, with $V(\mathfrak{a}_i) \subset X - V$. Blow up the product of the $\mathfrak{a}_i$. The blow-up $q : \widehat{X} \rightarrow X$ thus obtained is such that $q^{-1}(X_i)$ dominates the blow-up $\widehat{X}_i$. If $X'_i = q^{-1}(X_i)$, there exists an open U'_i of X'_i , with $U'_i \supset q^{-1}(U_i)$, such that, above X_i , $f(U_i)$ is a section of $U'_i \rightarrow Y$. One concludes that, after this blow-up,

$$f(U) = \bigcup_i f_i(U'_i).$$

For the proof of Theorem 1.2, we may replace Y by $f(U)$, and conclude by Case 3 and the following lemma. $\square$

Lemma (1.3). *The scheme $f(U)$ is quasi-affine over $\widehat{X}$.*

Proof. Indeed, $f(U)$ is quasi-finite over $\widehat{X}$. $\square$

Corollary (1.4, a variant of Chow's lemma). *Let $f : X \rightarrow S$ be a separated morphism of finite type. Then for every open subscheme U which is quasi-projective over S and dense in X , there exists a diagram of S -schemes*

$$\begin{array}{ccc} & X' & \\ q \swarrow & & \searrow j \\ X & & \overline{X}, \end{array}$$

where q is the blow-up morphism defined by an ideal $\mathfrak{a}$ satisfying $V(\mathfrak{a}) \subset X - U$, and where $\overline{X}$ is projective over S . In particular, q is projective and surjective and $q^{-1}(U) \simeq U$.

Proof. Let U^* be a projective S -scheme containing U as a schematically dense open subscheme. By Theorem 1.2, applied to the injection of U into X , one may choose U^* quasi-dominant over X . Thus one has a diagram

$$\begin{array}{ccc} U & \hookrightarrow & V \\ \downarrow & & \downarrow \varphi \\ U^* & \longrightarrow & X, \end{array} \quad (1.4.1)$$

and here, since U^* is proper over S , φ is proper. $\square$

Lemma (1.5). *Let*

$$\begin{array}{ccc} U & \hookrightarrow & V \\ \downarrow & & \downarrow \varphi \\ Y & \longrightarrow & X \end{array}$$

be a diagram of separated S -schemes of finite type, with U dense in X , dense in Y , and schematically dense in V , and with φ a quasi-domination of Y in X . If $\mathfrak{a}$ is an ideal of X , with $V(\mathfrak{a}) \subset X - U$, such that the blow-up $\widehat{X}$ of X along $\mathfrak{a}$ quasi-dominates V , relative to U , and such that U is schematically dense in $\widehat{X}$, giving a diagram

$$\begin{array}{ccc} U & \hookrightarrow & W \\ \downarrow & & \downarrow \psi \\ \widehat{X} & \longrightarrow & V, \end{array}$$

then:

(a) ψ is the blow-up morphism of V along $\varphi^*\mathfrak{a}$;

(b) the graph of ψ is closed in $\widehat{X} \times_S Y$.

Proof. Let $\widehat{V}$ be the blow-up of V along $\varphi^*\mathfrak{a}$. By the universal property of $\widehat{X}$, there is a unique X -morphism $\alpha : \widehat{V} \rightarrow \widehat{X}$. By the universal property of $\widehat{V}$, there is a unique V -morphism $\beta : W \rightarrow \widehat{V}$. The relevant diagram is

$$\begin{array}{ccccc} U & \hookrightarrow & W & \xrightarrow{\beta} & \widehat{V} \\ \downarrow & & \downarrow \psi & & \downarrow \alpha \\ V & \xleftarrow{\varphi} & Y & \longrightarrow & \widehat{X} \end{array}$$

The graph of φ is, trivially, proper over V ; the same is true of its inverse image on $V \times_S \widehat{X}$, and it follows that the graph of ψ is proper over V , hence that ψ is proper. Thus β is proper,

and its image is schematically dense, because already U is schematically dense in $\widehat{V}$. On the other hand, β is locally a section of α on W (because $\beta|_U$ is such a section over U); hence β is an isomorphism. By hypothesis the graph of ψ is closed in $V \times_S \widehat{X}$, and is contained in the inverse image of the graph of φ , which is closed in $Y \times X$. One concludes that $\Gamma(\psi)$ is closed in $Y \times \widehat{X}$. $\square$

Lemma (1.7). *Let*

$$X_2 \leftarrow U \rightarrow X_1$$

be a diagram of separated S -schemes of finite type, where U is a dense open subscheme of each X_i . Then there exists $j : U \hookrightarrow X$, with X separated of finite type over S and U schematically dense in X , and proper quasi-dominations

$$p_i : X \dashrightarrow X_i.$$

Proof. By Theorem 1.2, after blowing up X_1 without touching U , one reduces to the case where X_1 quasi-dominates X_2 :

$$\begin{array}{ccc} & X_1 & \\ \swarrow & & \searrow \\ U & \xhookrightarrow{\quad} & X_2. \end{array}$$

Apply Lemma 1.5 to this diagram, in order to obtain

$$\begin{array}{ccc} W & \xrightarrow{\psi} & X'_2 \\ \uparrow & & \uparrow \\ U & \xhookrightarrow{\quad} & V, \\ & & \uparrow \\ & & X_1 \end{array}$$

where

- (a) ψ is the blow-up morphism of an ideal $\mathfrak{a}$ of V , with $V(\mathfrak{a}) \subset V - U$;
- (b) the graph of ψ is closed in $X_1 \times X'_2$;
- (c) X'_2 is a blow-up of X_2 .

Let $\mathfrak{a}'$ be an ideal of X_1 extending $\mathfrak{a}$, with $V(\mathfrak{a}') \subset X_1 - U$, and let $q : X'_1 \rightarrow X_1$ be the blow-up of X_1 along $\mathfrak{a}'$. Then ψ induces an isomorphism between W and $q^{-1}(V)$, and the graph of this isomorphism is closed in $X'_1 \times X'_2$. Define X by gluing X'_1 and X'_2 along this isomorphism. $\square$

Remark. *The proof shows that one may take for $p_i : U_i \rightarrow X_i$, with $U_i \subset X$, a blow-up morphism of an ideal outside U .*

Theorem (1.6). *Let $f : X \rightarrow S$ be separated of finite type. There exists an immersion of X into an S -scheme proper over S .*

Proof. Let (U_i) be a finite open covering of X by dense quasi-projective open subschemes. For each U_i , there is, by Corollary 1.4, a commutative diagram of S -schemes

$$\begin{array}{ccc} X'_i & \xrightarrow{q_i} & X_i \\ j_i \downarrow & & \\ U_i & \xhookrightarrow{\quad} & \overline{X}_i, \end{array}$$

where q_i is proper, $\overline{X}_i$ is proper over S , and U_i is dense in $\overline{X}_i$. Let $F_i = \overline{X}_i - U_i$ and $F'_i = q_i^{-1}(F_i)$. Finally, let X^* be the closure of U in the product of the $\overline{X}_i$, with projections

$$p_i : X^* \rightarrow \overline{X}_i.$$

By Lemma 0.5, a preliminary blow-up of the $\overline{X}_i$, not touching X'_i , allows one to suppose that

$$\bigcap_i p_i^{-1}(F'_i) = \emptyset.$$

Let M_i be the S -scheme, separated over S , obtained by gluing X and $\overline{X}_i - F'_i$ along U_i :

$$\begin{array}{ccc} \overline{X}_i - F'_i & & X \\ & \swarrow \quad \searrow & \\ & M_i & \end{array}$$

Apply Lemma 1.7 to X and to the M_i , to obtain $X \hookrightarrow M$, where M properly quasi-dominates each M_i . It remains to prove that M is proper over S .

Indeed, one has morphisms

$$X^* - p_i^{-1}(F'_i) \longrightarrow M_i$$

which glue on $U = \bigcap_i U_i$. Replacing X^* by a blow-up X^{**} , one obtains morphisms

$$X^{**} - p_i^{-1}(F_i) \longrightarrow M$$

which glue, and hence a morphism from the schematic closure of U in X^{**} to M . The scheme M , as the image of this morphism, is proper over S . $\square$

Comments. *I hope that, by being still more constructive, one can prove Theorem 1.6 for $f : X \rightarrow S$ separated of finite type, with S quasi-compact and quasi-separated. The rules to follow are:*

- (a) *consider only quasi-compact open subsets and closed subsets defined by ideals of finite type;*
- (b) *never take schematic closure (one would leave (a)), but take only a closed subset containing the given part and sufficiently small for the later constructions;*
- (c) *do not speak of a dense quasi-projective open subset, cf. (b). Remember that, to construct such open subsets, one could start from an affine covering (U_i) , and take*

$$U = \bigcup_i \left(U_i - \bigcup_{j < i} U_j \right).$$

Translate the arguments in terms of the original U_i , and apply (a) and (b).

On the other hand, I have no idea in the case of algebraic spaces.

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References

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